WORKED SOLUTION

STEP I 1998, Question 5

Pure • Complex Numbers and Rotation

STEP I 1998, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s1-q5

The key idea

Both parts are about using the algebra of complex numbers to do the work for us. In part (i), going from one vertex of a regular hexagon to the next is a rotation about the centre through 60°, and a rotation about the origin is just multiplication by a complex number of modulus 1, so we work relative to Q. In part (ii), a number is real exactly when its imaginary part is zero, so we realise each denominator and set the imaginary part to zero.

Solution

Part (i)

Work relative to the centre. The vector from Q to A is

$$A - Q = (10 + 2i) - (4 + 6i) = 6 - 4i.$$

The six vertices of a regular hexagon are equally spaced round a circle centred on Q, so the angle AQB is $360^\circ/6 = 60^\circ$, and QB has the same length as QA. Moving *clockwise* means turning through -60°, that is, multiplying by

$$\mathrm{e}^{-i\pi/3} = \cos\tfrac{\pi}{3} - i\sin\tfrac{\pi}{3} = \tfrac{1}{2} - \tfrac{\sqrt{3}}{2}\,i.$$

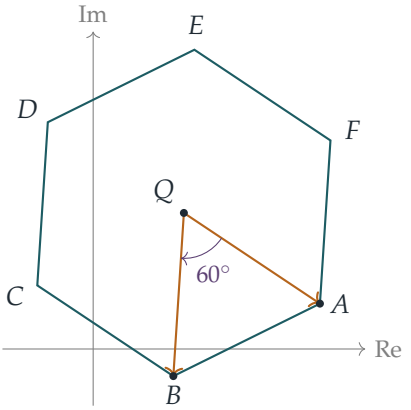
Ask yourself Why multiply $A - Q$ rather than A itself?

Multiplying by $\mathrm{e}^{-i\pi/3}$ rotates about the *origin*. To rotate about Q we shift Q to the origin (subtract Q), rotate, and shift back (add Q). So

$$\begin{aligned} B - Q &= (6 - 4i)\left(\tfrac{1}{2} - \tfrac{\sqrt{3}}{2}\,i\right) \\ &= 3 - 3\sqrt{3}\,i - 2i + 2\sqrt{3}\,i^2 \\ &= (3 - 2\sqrt{3}) - (2 + 3\sqrt{3})\,i, \end{aligned}$$

and adding $Q = 4 + 6i$,

$$\boxed{B = (7 - 2\sqrt{3}) + (4 - 3\sqrt{3})\,i} \approx 3.536 - 1.196\,i.$$



The picture confirms the direction: going clockwise from A, which lies down and to the right of Q, the next vertex lies almost directly below Q, as our B does.

Part (ii)

Realise each denominator by multiplying top and bottom by the conjugate, using $(1 + ki)(1 - ki) = 1 + k^2$:

$$\frac{1}{1+i} = \frac{1-i}{2}, \qquad \frac{1}{1+2i} = \frac{1-2i}{5}, \qquad \frac{1}{1+3i} = \frac{1-3i}{10}.$$

Since a, b, c are real, the imaginary part of the whole expression is

$$-\frac{a}{2} - \frac{2b}{5} - \frac{3c}{10} = -\frac{5a + 4b + 3c}{10}.$$

The expression is real exactly when this imaginary part is zero, so the condition is

$$\boxed{5a + 4b + 3c = 0}.$$

(Any non-zero integer multiple, such as $-5a - 4b - 3c = 0$, is the same condition.) In fact this condition is both sufficient and necessary, since we found the imaginary part exactly.

Checks

- In part (i), $|A - Q| = |6 - 4i| = \sqrt{52} = 2\sqrt{13}$. Our B gives $|B - Q|^2 = (3 - 2\sqrt{3})^2 + (2 + 3\sqrt{3})^2 = 21 - 12\sqrt{3} + 31 + 12\sqrt{3} = 52$, so B is on the right circle, and numerically $|B - A| \approx 7.211 = |A - Q|$ too, as it must be for an equilateral triangle QAB.

- In part (ii), try $a = 3$, $b = 0$, $c = -5$ (which satisfies $5a + 4b + 3c = 0$): $\frac{3(1-i)}{2} - \frac{5(1-3i)}{10} = \frac{3}{2} - \frac{1}{2} + \left(-\frac{3}{2} + \frac{3}{2}\right) i = 1$, which is real.
- A numerical check over 10 000 random triples with $c = -(5a + 4b)/3$ gave an imaginary part of zero every time.

Common slips

- Rotating anticlockwise (multiplying by $e^{i\pi/3}$) gives $(7 + 2\sqrt{3}) + (4 + 3\sqrt{3})i$, which is vertex F, not B.
- Rotating A about the origin instead of about Q.
- In part (ii), setting the *real* part to zero, or forgetting that a, b, c must be real for the imaginary part to be read off like this.

Beyond STEP

Multiplication by $e^{i\theta}$ is a rotation, and a map $z \mapsto \alpha z + \beta$ is a rotation and enlargement followed by a translation; these are the simplest examples of the Möbius transformations studied in a first course on complex analysis. The fact that the vertices of a regular polygon centred at the origin are z_0 times the roots of unity is the starting point for a great deal of algebra and geometry. Needham's *Visual Complex Analysis* develops this geometric view of complex multiplication beautifully.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.